Decision Tree Regression

Regression example • Jupyter Notebook-style code and output

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, r2_score
```

In [2]:

```
X, y = make_regression(n_samples=180, n_features=1,
                       noise=18, random_state=11)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

Out [ ]:

```
Training samples: 135
Test samples: 45
```

In [3]:

```
# Train a decision tree regressor
model = DecisionTreeRegressor(max_depth=3, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

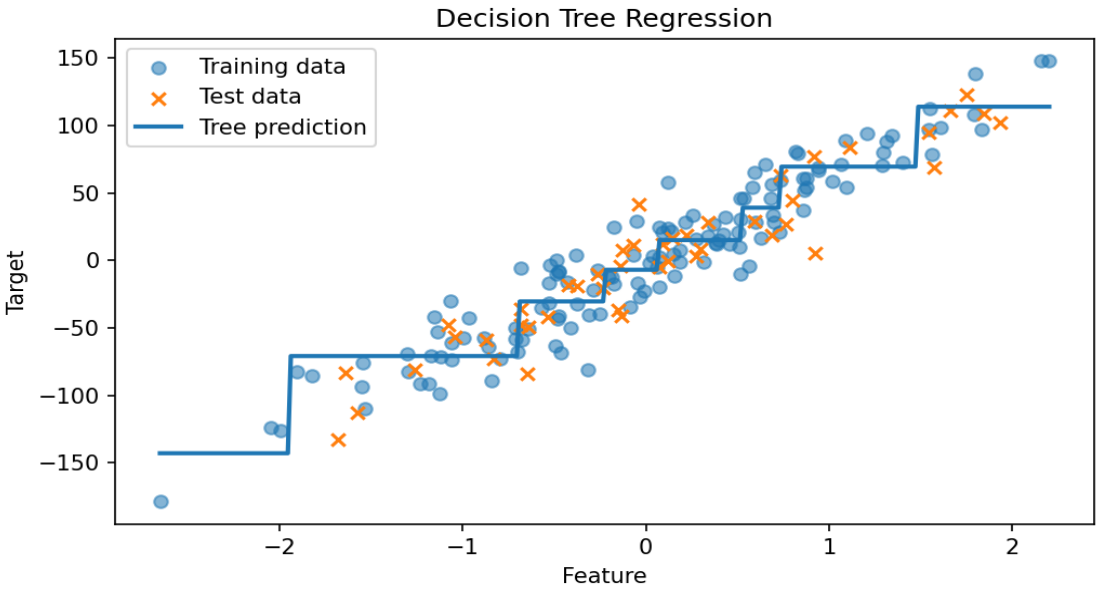
print("MAE:", round(mean_absolute_error(y_test, y_pred), 2))
print("R² score:", round(r2_score(y_test, y_pred), 3))
```

Out [ ]:

```
MAE: 18.77
R² score: 0.835
```

In [4]:

```
# Plot actual data and model predictions
x_line = np.linspace(X.min(), X.max(), 300).reshape(-1, 1)
plt.scatter(X_train[:, 0], y_train, alpha=0.55, label='Training data')
plt.scatter(X_test[:, 0], y_test, marker='x', label='Test data')
plt.plot(x_line[:, 0], model.predict(x_line),
         linewidth=2, label='Tree prediction')
plt.xlabel('Feature')
plt.ylabel('Target')
plt.title('Decision Tree Regression')
plt.legend()
plt.show()
```



Rendered output of the plotting cell